Jie Li

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RESEARCH INTERESTS

My research focuses on high-performance computing (HPC) and AI infrastructure, with an emphasis on observability, resource management, and energy efficiency.

- **High-performance computing:** system monitoring, workload characterization, scheduling, disaggregated memory, and energy-aware resource management.
- **AI infrastructure:** LLM serving, KV cache and memory management, and performance and power characterization of AI workloads.
- **AI agents for HPC operations and security:** agent-assisted cluster operation and management, including ongoing development for the NSF REPACSS system, and evaluating and constraining agent behavior in shared computing environments.
- **HPC and quantum computing (emerging):** quantum arithmetic and polynomial synthesis, with broader interests in hybrid quantum-classical workflows and HPC integration.

EDUCATION

Doctor of Philosophy, Computer Science, *Texas Tech University, Lubbock, TX* May 2024

- Dissertation: *Optimizing High-Performance Computing Systems: Insights from System Monitoring, Workload Management, and Scheduling Strategies*. Advisor: Prof. Yong Chen.

Master of Science, Computer Science, *Texas Tech University, Lubbock, TX* August 2019

- Thesis: *PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations*. Advisor: Prof. Yong Chen.

Bachelor of Arts, Architecture, *Huaqiao University, Xiamen, China* May 2012

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Research Assistant Professor July 2026 – Present

Department of Computer Science, Texas Tech University, Lubbock, TX

- Lead research in HPC and AI infrastructure, spanning observability, resource and data management, and energy-aware computing.
- Initiated a research direction on AI agents for HPC operation and management, studying how agents can assist with cluster tasks under scoped authorization and operator oversight, with REPACSS as the production testbed.
- Contribute to the operation, optimization, and research use of the NSF REPACSS cluster, including system software integration, performance analysis, and user support.
- Supervise graduate and undergraduate researchers.
- PI on AWARE and Co-PI on CACTUS and ASPIRE, all submitted to NSF in 2026 (see Grants and Proposals).

Assistant Director January 2026 – Present

NSF Cloud and Autonomic Computing Center (CAC IUCRC), Texas Tech Site

- Coordinate faculty, student teams, industry members, and external collaborators in developing and reviewing research projects for this NSF Industry–University Cooperative Research Center.
- Help organize the semiannual Industry Advisory Board meetings, including project planning and follow-up with industry participants, and support center reporting, proposal development, and cross-institutional coordination.
- Mentor students preparing presentations, posters, and reports for industry review.

Postdoctoral Researcher June 2024 – June 2026

Department of Computer Science, Texas Tech University, Lubbock, TX

- Co-designed and built the NSF REPACSS cluster, contributing to its system architecture, deployment, operation, and performance optimization.
- Served as technical lead of the Data-Intensive Scalable Computing Laboratory, coordinating research activities, student projects, and collaborative system development.

Research Assistant September 2019 – May 2024

Data-Intensive Scalable Computing Laboratory, Texas Tech University, Lubbock, TX

- Across graduate study (2019–2024), published 15 peer-reviewed papers (7 first-author) in venues including CLUSTER, ISC, ICPP, MemSys, IEEE CLOUD, IEEE BigData, and IEEE Transactions on Computers.
- Developed and maintained MonSTer, the Disaggregation-Aware Scheduler, and xBGAS simulation tools, supporting research in HPC monitoring, workload characterization, scheduling, and memory systems.
- Mentored students on HPC monitoring, data management, and workload analysis, including a master's thesis that led to a CLOUD'23 publication.

Graduate Student Intern Summers 2021, 2022, and 2023

Lawrence Berkeley National Laboratory, Berkeley, CA (Mentors: Brandon Cook and Georgios Michelogiannakis, in John Shalf's group)

- Designed and implemented pipelines integrating LDMS, DCGM, and Slurm telemetry from NERSC's Cori and Perlmutter supercomputers.
- Applied machine learning and deep learning to classify, characterize, and predict HPC job behavior from large-scale time-series telemetry.
- Led development of the open-source Disaggregation-Aware Scheduler; internship research resulted in first-author publications at ISC'23 and CLUSTER'24.

GRANTS AND PROPOSALS

Submitted Proposals (PI or Co-PI)

AWARE: Agent-Ready Workflow and Run-Record Enhancements for Weather Simulation Data 2026

NSF AIDatasets Planning · Submitted · Role: PI · Requested: $116,097

- Co-PIs: Yong Chen and Brian Ancell. Led proposal development for a one-year community planning effort to prepare weather simulation datasets for reliable use by AI agents, with workshops to define data requirements, provenance, governance, and an implementation roadmap.

CACTUS: Coupled AI-Ready Cyberinfrastructure Telemetry for Unlocking Scientific Discovery in Power-Aware Computing 2026

NSF AIDatasets: Impact · Submitted · Role: Co-PI · Requested: $1,706,677

- PI: Yong Chen; Co-PIs: Alan Sill, Argenis Bilbao, and Tara Salman. Responsible for data integration pipelines, AI-assisted metadata mapping, and benchmark development for an AI-ready dataset linking computing workloads, facility operations, energy supply, and weather.

ASPIRE: Advancing Scientific Discovery through High-Performance Computing and AI Research 2026

NSF REU Site · Submitted · Role: Co-PI · Requested: $430,000

- PI: Yong Chen; Co-PI: Maaz Amjad. Designed the undergraduate research projects in HPC and AI.

Contributions to Funded Projects

REPACSS: Empowering Scientific Discovery through Renewable Energy Powered Advanced Computing Systems and Services 2023

NSF Category II · Funded · Role: Contributor to proposal development and infrastructure implementation · Total project award: $12,250,000

- PI: Yong Chen; Co-PI: Alan Sill. Contributed proposal sections on data center monitoring and remote control; later co-designed and built the cluster as a postdoctoral researcher.

PEER-REVIEWED PUBLICATIONS

[19] R. Kadel, **J. Li**, and Y. Chen, "INSIGHT-HPC: An Interpretable System Intelligence for Global HPC Telemetry," in *2026 IEEE International Conference on Big Data (BigData'26)*, 2026. Accepted.

[18] Y. Yang, M. A. Rafi, X. Wang, K. Bouasker, **J. Li**, D. Wong, D. Li, and H. Jeon, "OCEAN: Open-Source CXL Emulation for Hyperscale Architecture and Networking," in *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'26)*, 2026. Accepted.

[17] Y. Zhao, **J. Li**, C. Niu, A. Sill, and Y. Chen, "Power-Centric Observability for HPC Systems: Design, Deployment, and Evaluation on REPACSS," in *Practice and Experience in Advanced Research Computing (PEARC'26)*, 2026, pp. 1–8.

[16] C. Niu, W. Zhang, **J. Li**, Y. Zhao, T. Wang, X. Wang, and Y. Chen, "TokenPowerBench: Benchmarking the Power Consumption of LLM Inference," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'26)*, 2026, pp. 32582–32590. doi: 10.1609/aaai.v40i38.40535. arXiv:2512.03024.

[15] **J. Li**, G. Michelogiannakis, S. Maloney, B. Cook, E. Suarez, J. Shalf, and Y. Chen, "Job Scheduling in High Performance Computing Systems with Disaggregated Memory Resources," in *2024 IEEE International Conference on Cluster Computing (CLUSTER'24)*, 2024, pp. 297–309. doi: 10.1109/CLUSTER59578.2024.00033.

[14] **J. Li**, G. Michelogiannakis, B. Cook, J. Shalf, and Y. Chen, "Scheduling and Allocation of Disaggregated Memory Resources in HPC Systems," in *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW'24)*, 2024, pp. 1202–1203. doi: 10.1109/IPDPSW63119.2024.00206.

[13] **J. Li**, J. D. Leidel, B. Page, and Y. Chen, "Towards Cycle-accurate Simulation of xBGAS," in *2024 International Conference on Computing, Networking and Communications (ICNC'24)*, 2024, pp. 468–472. doi: 10.1109/ICNC59896.2024.10556078.

[12] T. Dang, N. V. T. Nguyen, **J. Li**, A. Sill, and Y. Chen, "Spiro: Order-Preserving Visualization in High Performance Computing Monitoring," in *International Symposium on Visual Computing (ISVC'23)*, 2023, pp. 109–120. doi: 10.1007/978-3-031-47969-4_9.

[11] C. E. Caon, **J. Li**, and Y. Chen, “Effective Management of Time Series Data,” in *2023 IEEE 16th International Conference on Cloud Computing (CLOUD’23)*, 2023, pp. 408–414. doi: 10.1109/CLOUD60044.2023.00055.

[10] **J. Li**, R. Wang, G. Ali, T. Dang, A. Sill, and Y. Chen, “Workload Failure Prediction for Data Centers,” in *2023 IEEE 16th International Conference on Cloud Computing (CLOUD’23)*, 2023, pp. 479–485. doi: 10.1109/CLOUD60044.2023.00064.

[9] **J. Li**, G. Michelogiannakis, B. Cook, D. Cooray, and Y. Chen, “Analyzing Resource Utilization in an HPC System: A Case Study of NERSC’s Perlmutter,” in *International Conference on High Performance Computing (ISC’23)*, 2023, pp. 297–316. doi: 10.1007/978-3-031-32041-5_16.

[8] T. Dang, N. V. T. Nguyen, **J. Li**, A. Sill, J. Hass, and Y. Chen, “JobViewer: Graph-based Visualization for Monitoring High-Performance Computing System,” in *2022 IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT’22)*, 2022, pp. 110–119. doi: 10.1109/BDCAT56447.2022.00021.

[7] T. Dang, N. Nguyen, J. Hass, **J. Li**, Y. Chen, and A. Sill, “The Gap between Visualization Research and Visualization Software in High-Performance Computing Center,” *The Gap between Visualization Research and Visualization Software (VisGap’21)*, 2021, doi: 10.2312/visgap.20211089.

[6] X. Wang, A. Tumeo, J. D. Leidel, **J. Li**, and Y. Chen, “HAM: Hotspot-Aware Manager for Improving Communications With 3D-Stacked Memory,” *IEEE Transactions on Computers (IEEE Trans Comput)*, vol. 70, no. 6, pp. 833–848, 2021, doi: 10.1109/TC.2021.3066982.

[5] **J. Li**, G. Ali, N. Nguyen, J. Hass, A. Sill, T. Dang, and Y. Chen, “MonSTer: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems,” in *2020 IEEE International Conference on Cluster Computing (CLUSTER’20)*, 2020, pp. 119–129. doi: 10.1109/CLUSTER49012.2020.00022.

[4] N. Nguyen, J. Hass, Y. Chen, **J. Li**, A. Sill, and T. Dang, “RadarViewer: Visualizing the Dynamics of Multivariate Data,” in *Practice and Experience in Advanced Research Computing (PEARC’20)*, 2020, pp. 555–556. doi: 10.1145/3311790.3404538.

[3] V. Pham, N. Nguyen, **J. Li**, J. Hass, Y. Chen, and T. Dang, “MTSAD: Multivariate Time Series Abnormality Detection and Visualization,” in *2019 IEEE International Conference on Big Data (BigData’19)*, 2019, pp. 3267–3276. doi: 10.1109/BigData47090.2019.9006559.

[2] **J. Li**, X. Wang, A. Tumeo, B. Williams, J. D. Leidel, and Y. Chen, “PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations,” in *Proceedings of the International Symposium on Memory Systems (MemSys’19)*, 2019, pp. 41–52. doi: 10.1145/3357526.3357550.

[1] X. Wang, A. Tumeo, J. D. Leidel, **J. Li**, and Y. Chen, “MAC: Memory Access Coalescer for 3D-Stacked Memory,” in *Proceedings of the 48th International Conference on Parallel Processing (ICPP’19)*, 2019, pp. 1–10. doi: 10.1145/3337821.3337867.

PREPRINTS AND MANUSCRIPTS

Preprints

[3] **J. Li**, “Trusted Credentials, Untrusted Behavior: Benchmarking LLM-Agent Security in High-Performance Computing,” *arXiv preprint arXiv:2607.18485*, 2026.

[2] **J. Li**, T. Wang, and Y. Chen, “From Tensor Buffer to Distributed Memory Hierarchy: A Survey of KV Cache Management for LLM Serving,” *arXiv preprint arXiv:2607.02574*, 2026.

[1] **J. Li**, B. Cook, and Y. Chen, "ARcode: HPC Application Recognition Through Image-encoded Monitoring Data," *arXiv preprint arXiv:2301.08612*, 2023, doi: 10.48550/arXiv.2301.08612.

Manuscripts under Submission or Revision

- Ziqing Guo, **Jie Li**, Ziwen Pan, and Yong Chen. "NNQA: Neural-Native Quantum Arithmetic for End-to-End Polynomial Synthesis." Submitted to AAAI-27.
- Tongyang Wang, **Jie Li**, Chenxu Niu, and Yong Chen. "ECHO: Online Control of CPU Frequency for Energy-Efficient High-Performance Computing."
- Batuhan Sencer, Chenxu Niu, **Jie Li**, and Yong Chen. "Slimming Models, Saving Watts: Understanding and Modeling the Impact of Knowledge Distillation on GPU Clusters."

TEACHING

Teaching Experience

Parallel Processing (graduate), *Texas Tech University* Fall 2025
Role: Invited lecturer and programming project designer

- Delivered four invited lectures; topics included job scheduling, compilation and job submission, and OpenMP.
- Designed programming projects for 27 students.

Teaching Interests

- **Established courses** (undergraduate and graduate): operating systems, parallel and high-performance computing, computer architecture, distributed and cloud computing, and systems for machine learning.
- **Proposed graduate electives**: *AI Infrastructure: Systems for Training and Serving Large Models* and *AI Agents for Computing Systems*.

RESEARCH MENTORING

Graduate Students

- **Jaechang Kim**, M.S. student. Research mentoring. AI agents for REPACSS cluster operation and management, including agent-assisted Warewulf, Ansible, Spack, and Slurm workflows. Summer 2026 – Present
- **Rupak Kadel**, Ph.D. student. Research mentoring. Improving HPC monitoring for anomaly and intrusion detection. Fall 2025 – Present
- **Cristiano Caon**, M.S. Independent study (CS 7000) and master's thesis research. Data volume reduction and query optimization in time-series databases. Outcome: CLOUD'23 paper, "Effective Management of Time Series Data." Fall 2022
- **Aniruddh Sanjaysinh Chavda** (M.S.) and **Huyen Nguyen** (Ph.D.). Course project, Advanced Operating Systems. Usage behavior analysis by clustering job accounting data. Spring 2021
- **Ruonan Wu**, M.S. Course project, Advanced Operating Systems. Job accounting data analysis for the Quanah cluster. Spring 2021
- **Ashhrita Puradamane Balachandra**, M.S. Course project, Advanced Operating Systems. Improving InfluxDB query performance. Spring 2020

Undergraduate Students

- **Yusheng Han** and **Zachary Kay**. Independent study (CS 4000). Running HPC applications and analyzing performance on the RedRaider cluster. Spring 2022
- **Casey Root**. Independent study (CS 4000). Monitoring queue status through the Slurm REST API. Spring 2021

OPEN-SOURCE RESEARCH SOFTWARE

- **MonSTer**: Out-of-the-box HPC monitoring framework. Published at CLUSTER'20; adopted by Dell's Omnia project.
- **Disaggregation-Aware Scheduler**: Simulation framework for job scheduling in HPC systems with disaggregated memory. Used in CLUSTER'24 and IPDPSW'24 publications.
- **xBGAS REV-CPU**: Cycle-accurate xBGAS simulation extension based on REV-CPU. Developed with Tactical Computing Laboratories; published at ICNC'24.
- **xBGAS Runtime**: Lightweight runtime supporting global address space extensions for RISC-V-based xBGAS simulation. Developed with Tactical Computing Laboratories.

PRESENTATIONS

- Integrated Data Collection and Visualization Framework for Data Centers based on Telemetry Model. *NSF CAC Industry Advisory Board Conferences (Lubbock, TX; Denton, TX; Tucson, AZ; Lubbock, TX), 2021–2024*
- Towards Cycle-Accurate Simulation of xBGAS. *Latch-Up 2024, Cambridge, MA*
- Workload Failure Prediction for Data Centers. *IEEE CLOUD'23, Chicago, IL*
- Advanced Visualization and Data Analysis of HPC Cluster and User Application Behavior. *ACM/IEEE SC'21*
- MonSTer: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems. *IEEE CLUSTER'20*
- PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations. *MemSys'19*

REFEREED POSTERS

- **J. Li**, B. Cook, G. Michelogiannakis, and Y. Chen. A Holistic View of Memory Utilization on Perlmutter. *SC'22*
- **J. Li**, B. Cook, and Y. Chen. Detecting and Identifying Applications by Job Signatures. *SC'21*
- X. Wang, **J. Li**, A. Tumeo, J. D. Leidel, and Y. Chen. Memory Hotspot Optimizations for 3D-Stacked Memory. *PACT'19*

AWARDS AND HONORS

- Best Poster Award, *NSF Cloud and Autonomic Computing Center Industry Advisory Board Conference* 2022
- Summer Thesis/Dissertation Research Award ($2,300), *Texas Tech University Graduate School* Summer 2019

PROFESSIONAL SERVICE

- Program committee: NeurIPS 2026; AAAI 2026.
- Conference reviewer: IEEE ISCAS 2026; IEEE BigData 2022, 2023, 2025; IEEE/ACM CCGrid 2024.
- Journal reviewer: IEEE Computer Architecture Letters (2025); The Journal of Supercomputing (2023).
- Conference sub-reviewer: IEEE IPDPS 2023, 2027; IEEE ICDCS 2022; ACM/IEEE SC 2022; International Parallel Data Systems Workshop (PDSW) 2022; IEEE Smart Data Services 2020.
- Student volunteer: ACM/IEEE SC'21, St. Louis, MO; ACM/IEEE SC'19, Denver, CO.